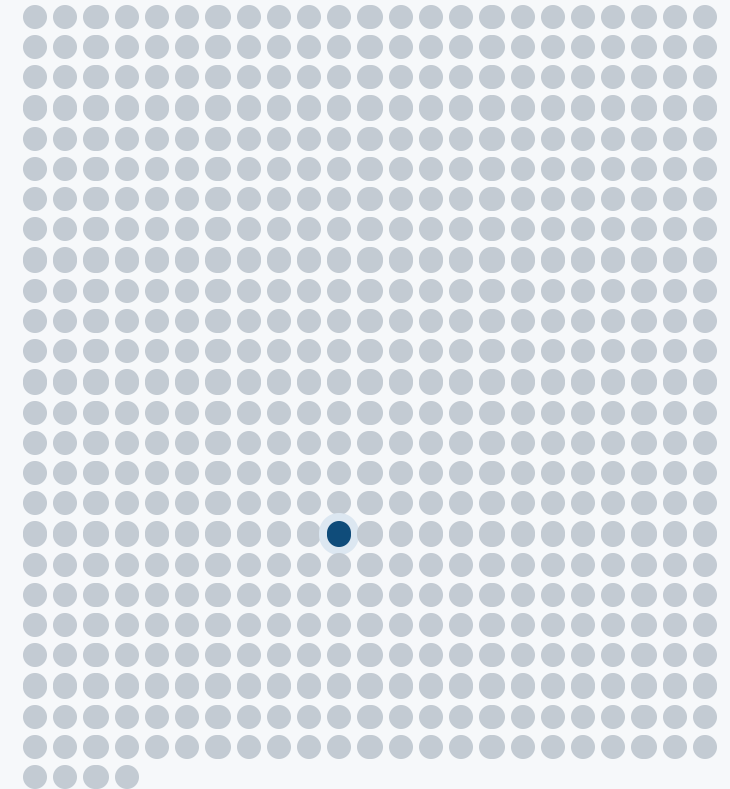


needle

Cost-aware credit card fraud
detection under extreme class
imbalance

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SECTION I

Introduction

0.17%

284,807 transactions

492 frauds

two days, one European issuer

A model that never predicts fraud is 99.83% accurate, and worth nothing.

ROC-AUC is nearly as uninformative as accuracy

Its abscissa divides false alarms by all 284,315 negatives. At 578 negatives for every positive, the entire span between a model an analyst can use and one that buries them fits inside the leftmost few percent of the curve.

Reported for comparability. Never used to choose.

SECTION II

Related work

Where the data comes from

PROVENANCE

Worldline and the Machine Learning Group at ULB

The extract is a by-product of an industrial research collaboration rather than a benchmark built to be topped.

THE FRAMING WE BORROW

Fraud detection is alert generation

The group treats the task as filling a bounded review queue, not as labelling every transaction. That is the framing this project adopts wholesale.

WHY WE CALIBRATE AT ALL

Undersampling shifts the posterior

Shown analytically by the same group, which is the reason calibration is a stage of this pipeline rather than an afterthought.

OUT OF REACH HERE

Two frozen days express none of it

The same line of work goes on to add verification latency and concept drift, streaming at scale, active learning under a labelling budget, domain adaptation and incremental retraining. A static extract carries none of it, so none of it is tested here.

What the literature already settles

II-B · EVALUATION

ROC and PR space weight the same errors differently

Established for classification in general, then argued directly for skewed data. The decision-theoretic basis for taking a threshold from costs rather than from a metric is older still.

II-C · RESAMPLING

SMOTE is expected to lose to class weighting

Its benefit largely vanishes in high dimensions where the minority class is not locally dense, and it rarely beats weighting for strong ensembles once the threshold is tuned.

II-D · LEAKAGE

Four failures recur in work on this dataset

- preprocessing applied before the split
- reporting too vague to reproduce
- no temporal validation at all
- recall bought at precision's expense

Found by a 2025 audit of published methodology on this dataset.

Every element of the protocol that follows answers one of those four, and it was fixed before any model was compared.

SECTION III

Methodology

What we optimise instead

- PR-AUC is the primary metric, because it answers to the minority class and so moves when the model does.
- Recall at precision ≥ 0.90 reports what a review team actually experiences at a tolerable rate of false alarms.
- An **amount-weighted cost** decides between the survivors, pricing every missed fraud at its own amount and every false alarm at one review.

$$\text{cost}(t) = \sum_{\text{missed frauds}} a_i + c_{\text{review}} \cdot |\text{false alarms}|$$

Here a_i is the amount of the transaction and $c_{\text{review}} = 3$ is the price of one analyst review. Because that price is an assumption rather than a measurement, every run re-solves the optimum over 1, 3, 5, 10, 30 and 100.

Leakage guards

BEFORE THE SPLIT

1,081 duplicate rows removed

19 of them frauds. An identical row landing on both sides of a split is leakage that no later care undoes.

THE SPLIT

Temporal, not random

Day 1 (144,236 rows, 272 frauds) for the search, the model selection and the threshold.

Day 2 (139,490 rows, 201 frauds) for one final evaluation, and nothing else.

INSIDE THE FOLDS

Scaling and resampling refitted per fold

Every candidate is an `imblearn.Pipeline`, so nothing is ever fitted on the fold being scored.

THE SPLITS INSIDE DAY 1

Five folds to search, fifteen to select

The hyperparameter search uses a stratified 5-fold, model selection a repeated stratified 5-fold with three repeats for fifteen folds in all, and the threshold is chosen from the out-of-fold scores alone.

Three feature decisions

- **V1** – **V28** are left untouched, because they already come out of a PCA and are therefore scaled and uncorrelated.
- **Amount** becomes $\log_amount = \log(1 + Amount)$ and then passes through a RobustScaler, since the tail is heavy and the outliers are the rows we care about.
- **Time** becomes $hour = (Time / 3600) \bmod 24$, because a raw second offset across two days is an index rather than a signal.

Seventeen candidates

- **Four learners:** stratified dummy, logistic regression, random forest and LightGBM, each of them a scikit-learn estimator wrapped in an imbalanced-learn pipeline.
- **Three imbalance strategies:** class weighting, random undersampling and SMOTE.
- **Two unsupervised detectors:** an isolation forest and a denoising autoencoder trained on legitimate rows only, to ask whether labels are needed at all.

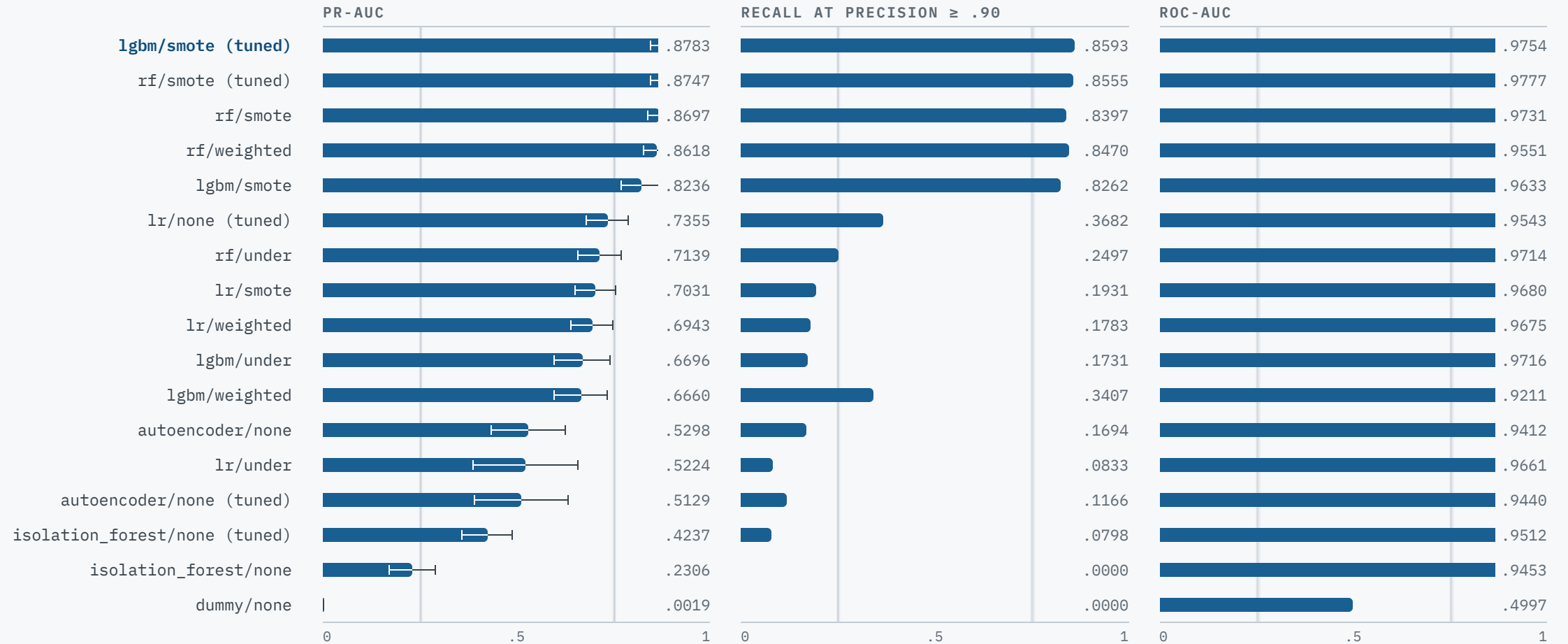
An Optuna search with a TPE sampler maximises PR-AUC over five folds, on a budget of 60 trials for LightGBM, which has the most to tune, and 15 to 20 for the others.

Every tuned winner is then re-scored on the full fifteen folds before it enters the leaderboard, to pay back the selection bias of having been chosen.

SECTION IV

Results

All seventeen candidates, ranked by PR-AUC



What the leaderboard says

The two metrics disagree on order, not just scale

ROC-AUC spans .9211–.9777; PR-AUC spans .2306–.8783. The isolation forest outscores weighted LightGBM on ROC-AUC with a third of its PR-AUC.

The imbalance strategy outweighs the learner

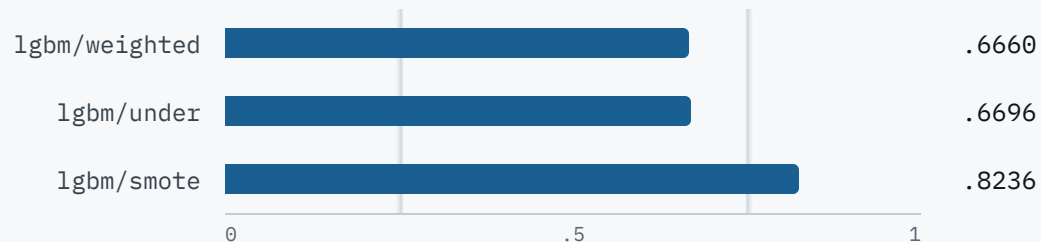
Within LightGBM alone the three strategies span 0.158 PR-AUC: .666 weighted, .670 under, .824 SMOTE. Wider than the gap between learners.

Labels are worth a lot

The best unsupervised detector reaches .5298 against the winner's .8783. Anomaly detection alone does not close it.

One result that contradicts the literature

PR-AUC BY IMBALANCE STRATEGY, LIGHTGBM

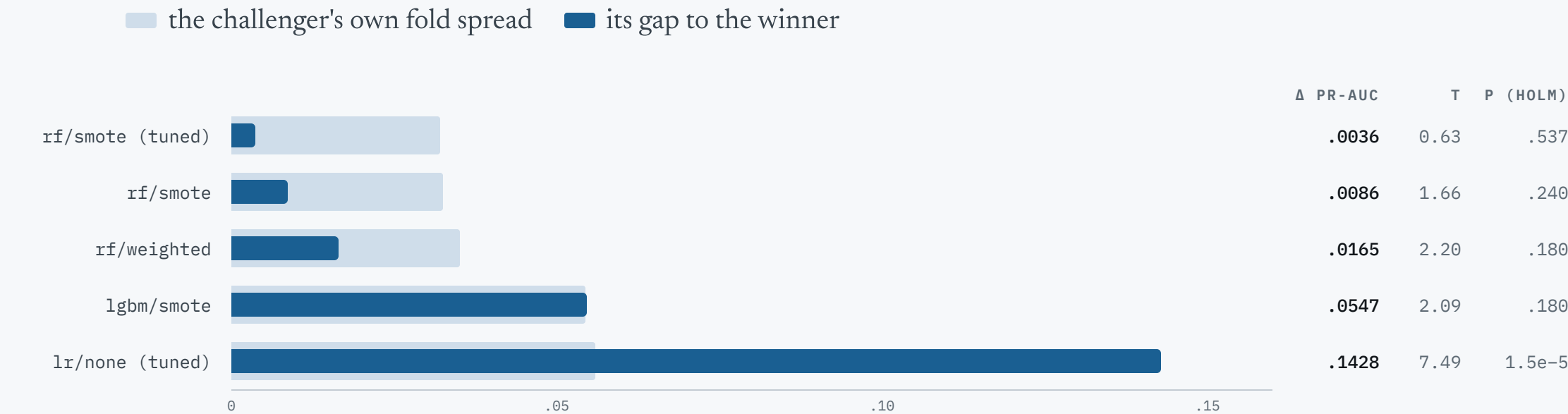


Section II-C said this should not happen: in high dimensions, for strong ensembles, with the threshold tuned, SMOTE is the strategy expected to lose.

Here it won for both tree ensembles, by a margin wider than anything that separates the learners themselves. The honest statement is narrower than that headline, though: SMOTE beat weighting on this data, at this seed, with the strategy tuned jointly with the learner. It is not a general result, and it is not offered as one.

Is the winner actually winning?

Corrected resampled *t*-test on 15 paired folds, Holm-adjusted, $\alpha = 0.05$ fixed in advance.



Four of the top five are indistinguishable

Each of those four gaps fits inside the spread between folds. Only the tuned logistic regression separates, by 0.14 PR-AUC against a fold spread of 0.035.

So the order of the top five rows is not supported by the data. What ships is the top of the leaderboard, a choice made before the test was run. The test does not overturn it; it says the four rows beneath would have served as well.

Day 2, touched once

PR-AUC

.878 → .764

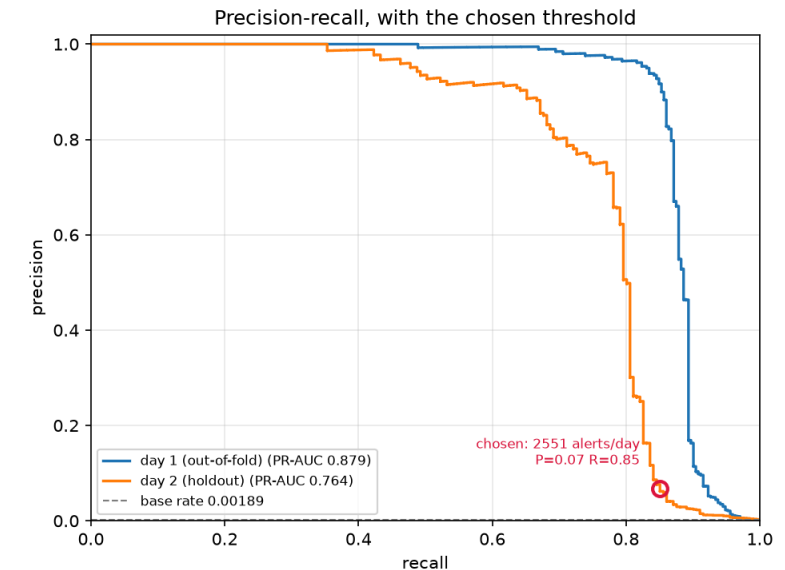
RECALL AT PRECISION .90

.859 → .652

ROC-AUC (IT RISES)

.9754 → .9790

- Random folds are too easy. Fraud arrives in bursts on the same stolen card, so two frauds are often near-identical rows and a random split puts one in train, one in test.
- The base rate moved. 0.189% on day 1 to 0.144% on day 2, a quarter fewer. PR-AUC depends on that share, so part of the drop is arithmetic.
- The rest is real drift. Two days give one split, so we cannot apportion the three.



SECTION IV-D

The operating point and its transfer

Three objectives, and what day 2 does to the one we shipped

OPERATING POINT	ALERTS	PRECISION	RECALL	COST
Day 1 00F · cost	524	.4637	.8934	4,234
Day 1 00F · precision floor .90	253	.9170	.8529	5,301
Day 1 00F · budget 100	100	1.000	.3676	22,217
Day 2 · same threshold	2,551	.0670	.8507	15,712
Day 2 · same alert rate	507	.3195	.8060	10,680
Day 2 · own optimum (hindsight)	217	.7235	.7811	9,871

COST ABOVE THE DAY-2 OPTIMUM IF THE THRESHOLD IS HELD

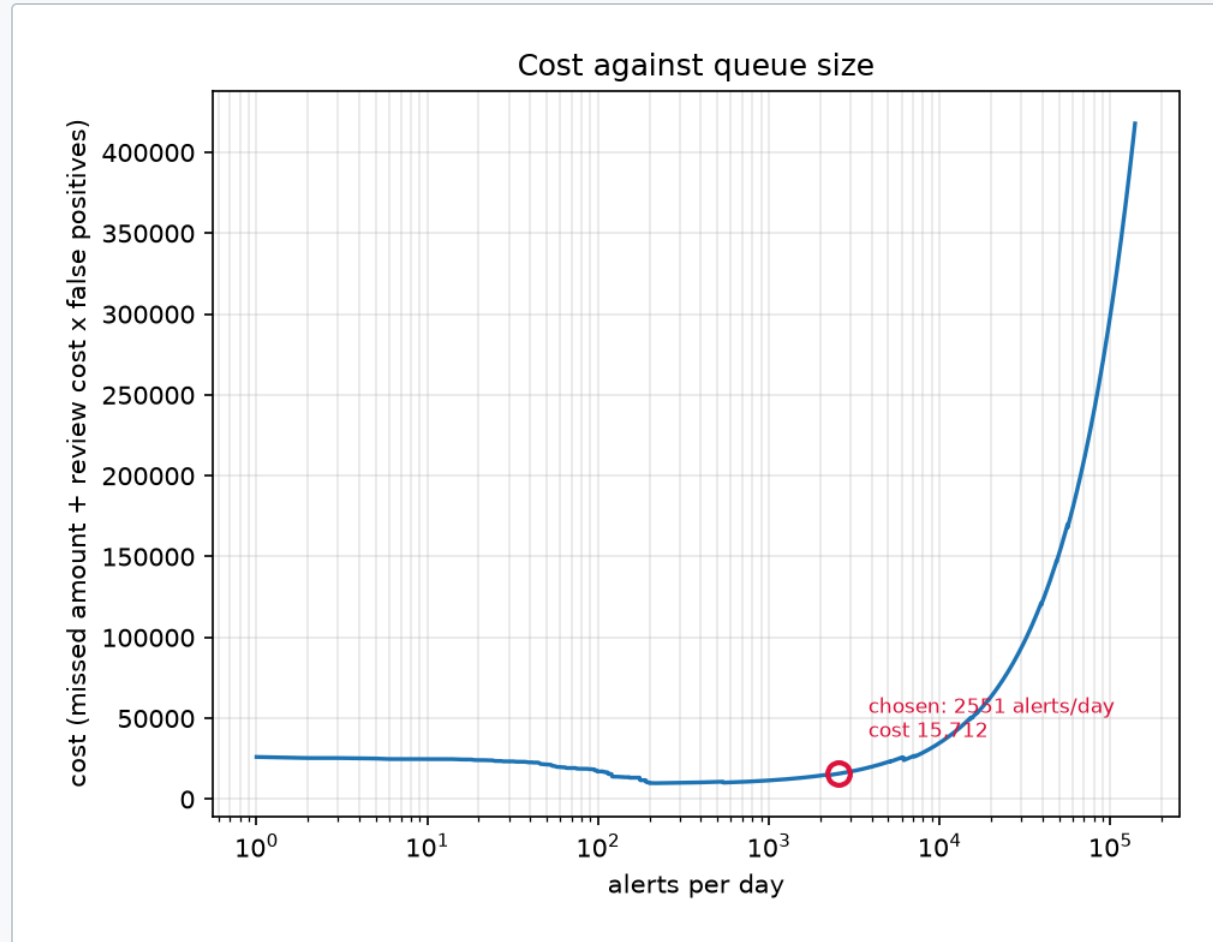
+59%

COST ABOVE THE DAY-2 OPTIMUM IF THE ALERT RATE IS HELD

+8%

Neither transfer is ideal, but one is seven times less wrong. Recall held up (.851 on day 2 at the frozen threshold) while precision fell from .46 to .07. The model kept finding fraud; the queue stopped being reviewable.

III-F THRESHOLD, CALIBRATION AND INTERPRETATION

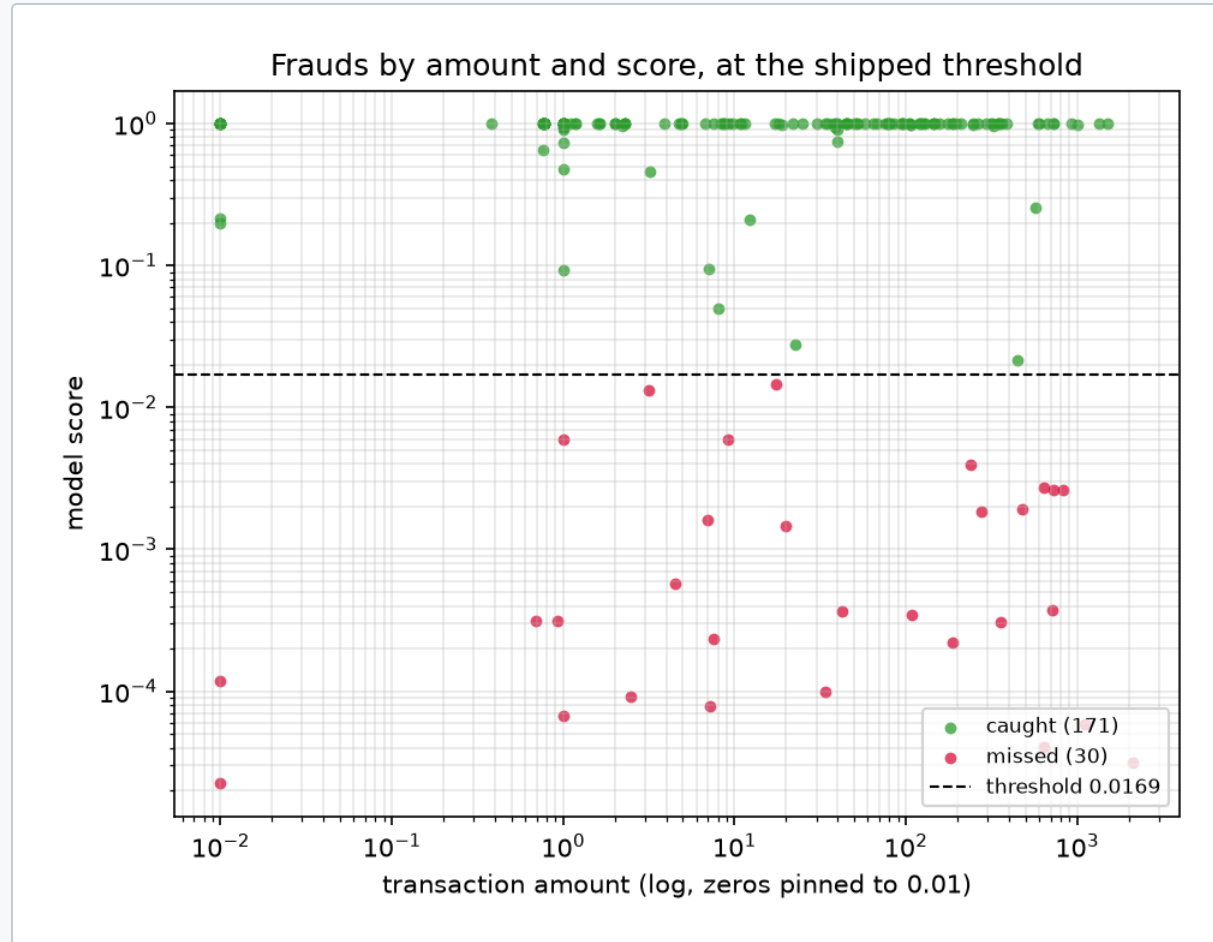


The price of a review is an assumption

Raising c_{review} a hundredfold moves the threshold from **0.0169** to **0.9919**, almost the whole interval.

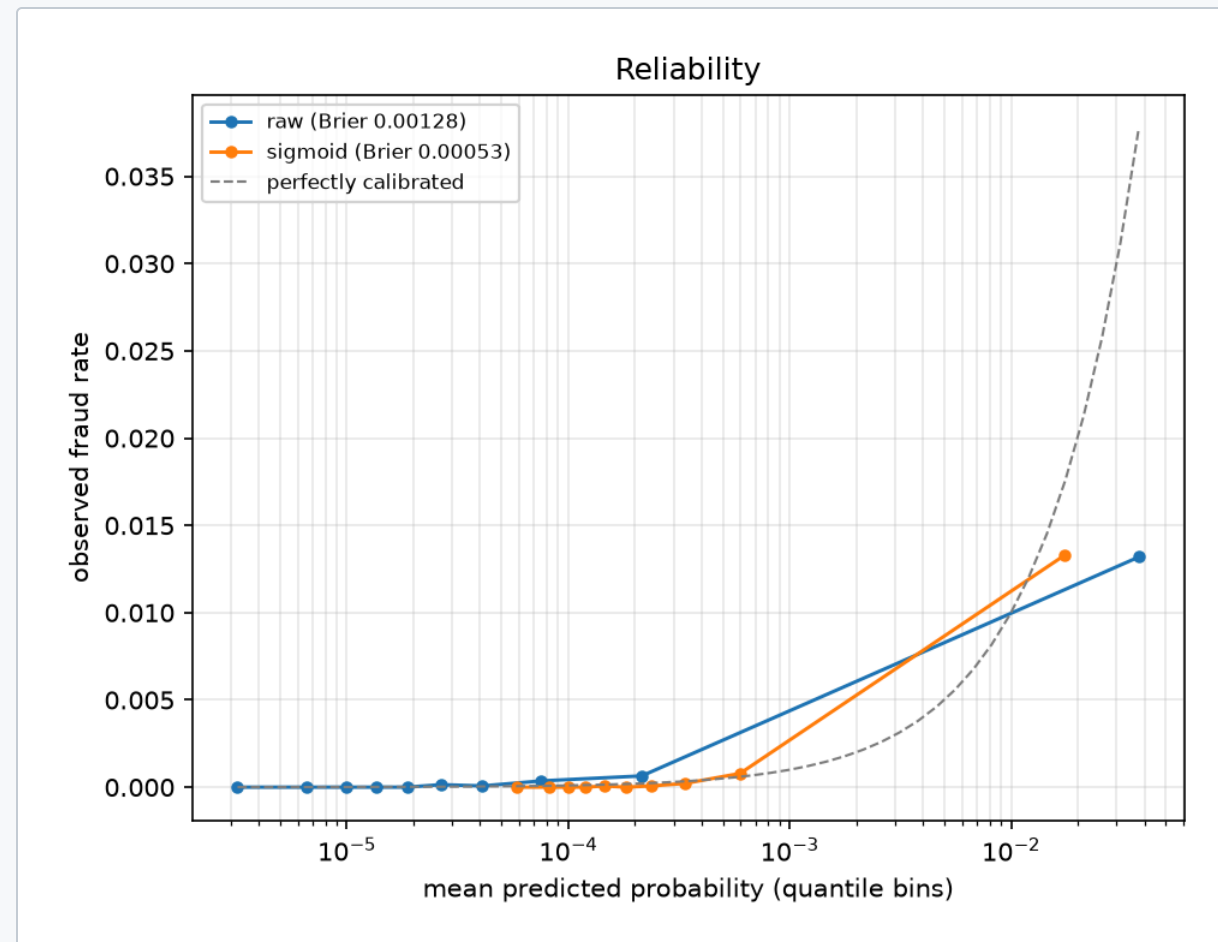
The queue only shrinks from **524** to **233** alerts, and recall from **.893** to **.824**.

IV-D THE OPERATING POINT AND ITS TRANSFER



Recall counts frauds, not money

Day-2 queue: 2,551 alerts, 171 of them fraud. The 30 frauds that get through are 14.9% of the count but 32.8% of the fraudulent amount.



A threshold needs order. A probability needs truth.

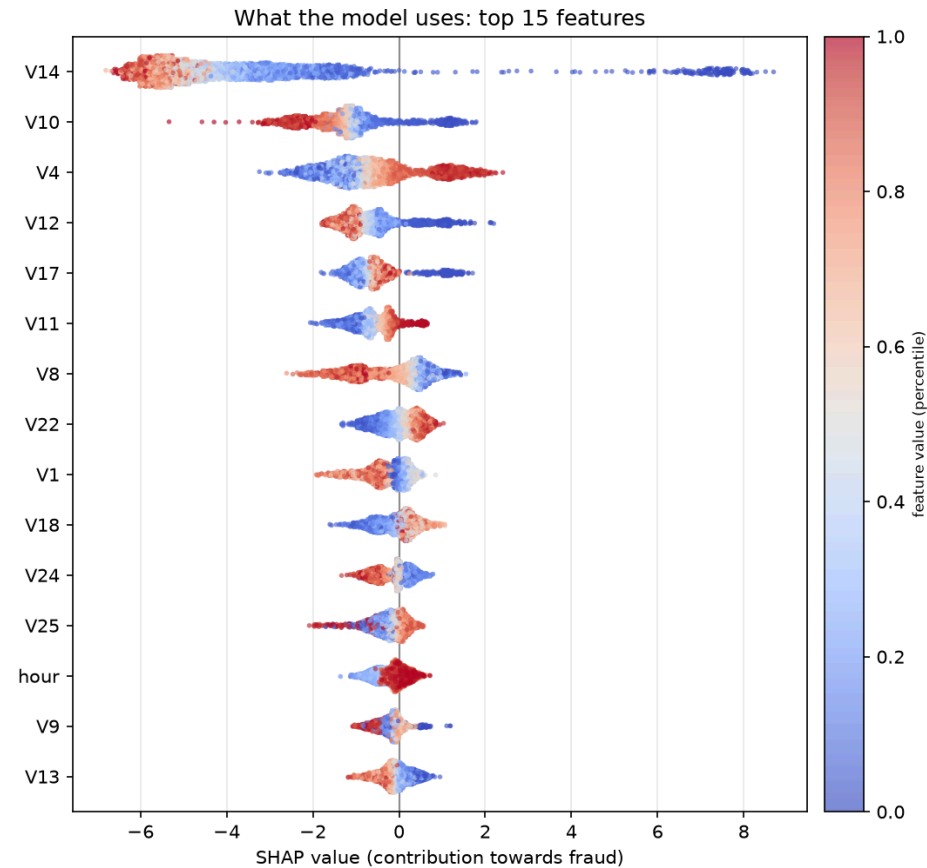
SMOTE trains the model on a world where fraud is common, so the raw scores overstate risk by $2.7\times$: mean score **.003821** against a real rate of **.001441**.

BRIER, RAW

.001283

BRIER, SIGMOID

.000526



The decision rests on a handful of features

SHAP with the exact tree explainer

V14 alone carries 33.5% of the total attribution; the top five carry 62.4%. Its low values push hardest towards fraud, further than any other feature.

Our own features earn a modest place: `hour` is 13th at 1.7%, `log_amount` 16th at 1.3%, below fourteen PCA components.

Frauds are deliberately over-represented in the sample, so the ranking answers "what separates the classes". And since the PCA was never published, we can say which features matter, not what they mean.

SECTION V

Discussion and limitations

Where the evidence stops

THE BINDING CONSTRAINT

473 frauds, 272 trainable

This is why the CV is repeated, why a fold spread is quoted with every mean, why the test needs a variance correction, and why calibration is sigmoid.

STRUCTURAL

Two days is one split

One drift measurement, no seasonality, no verification latency: a fraud is labelled here the instant it happens, which is not how it works.

ASSUMED, NOT MEASURED

The cost model

A miss priced at the full amount, a review at a flat 3. No recovery, no chargeback fees, no cost for a wrongly-blocked customer.

UNDIAGNOSABLE

Anonymised features

A PCA that was never published rules out every domain check, so the expensive misses cannot be explained, only counted.

UNTESTED

The unsupervised branch

The literature finds value in *combining* outlier scores with a supervised model rather than choosing between them. We chose.

Three things to take away

- At a 0.167% positive rate, ROC-AUC cannot separate a deployable **ranker from an unusable one**. It varied eleven times less than PR-AUC across the sixteen real candidates, inverted their order outright, and improved across the split on which precision fell sevenfold.
- **The winner is a winner by leaderboard order, not by evidence.** Tuned LightGBM over SMOTE ($.878 \pm .035$ in CV, .764 on the holdout), but the corrected paired test cannot separate it from four challengers, where the uncorrected test would have separated four. The resampling strategy proved worth more than the learner.
- **Specify alert volume, not a probability cut-off.** Carried to day 2 the frozen threshold held recall and dropped precision from .46 to .07, raising 2,551 alerts against day 2's own optimum of 217.

Thank you!

github.com/ITHackerstein/needle